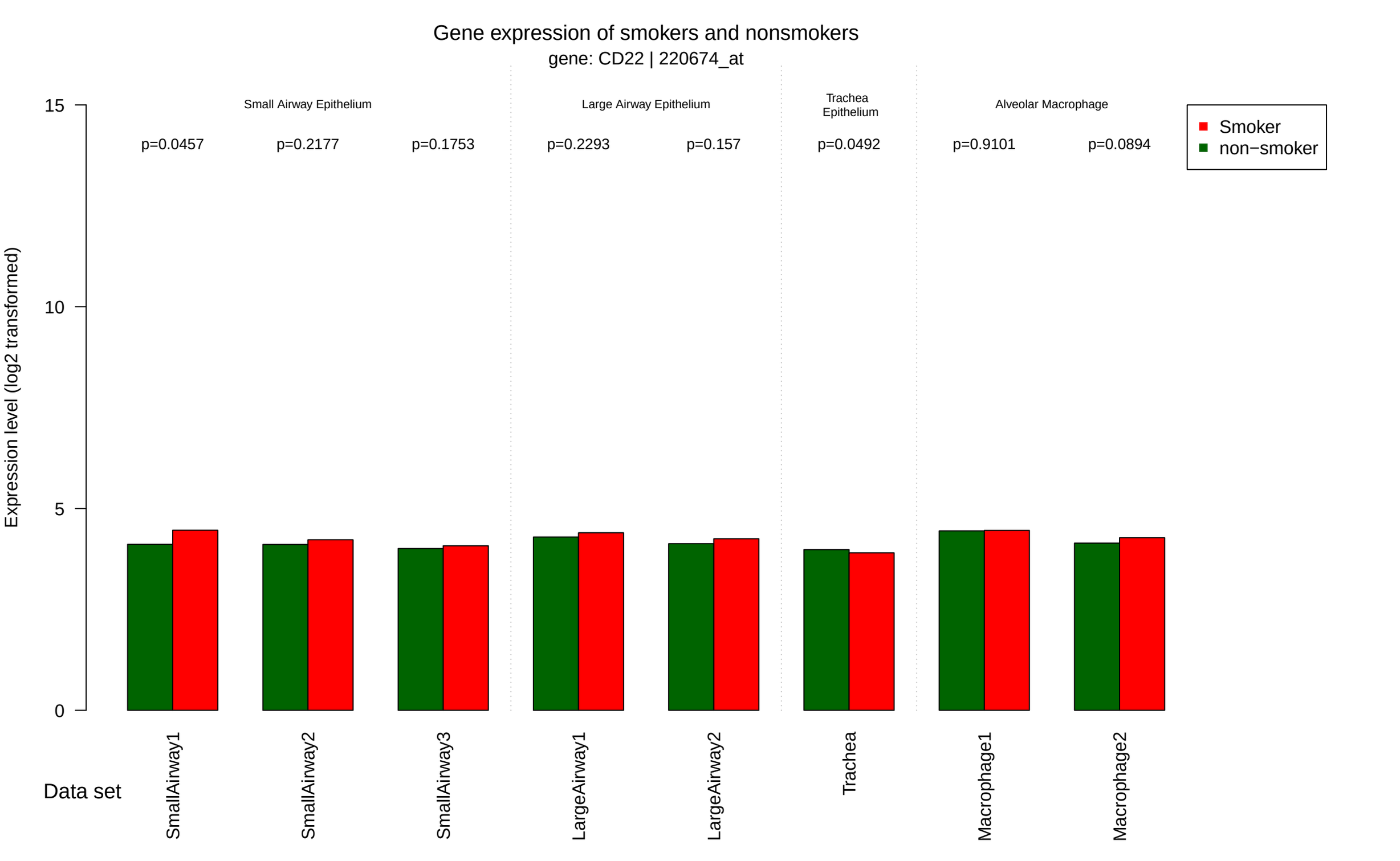
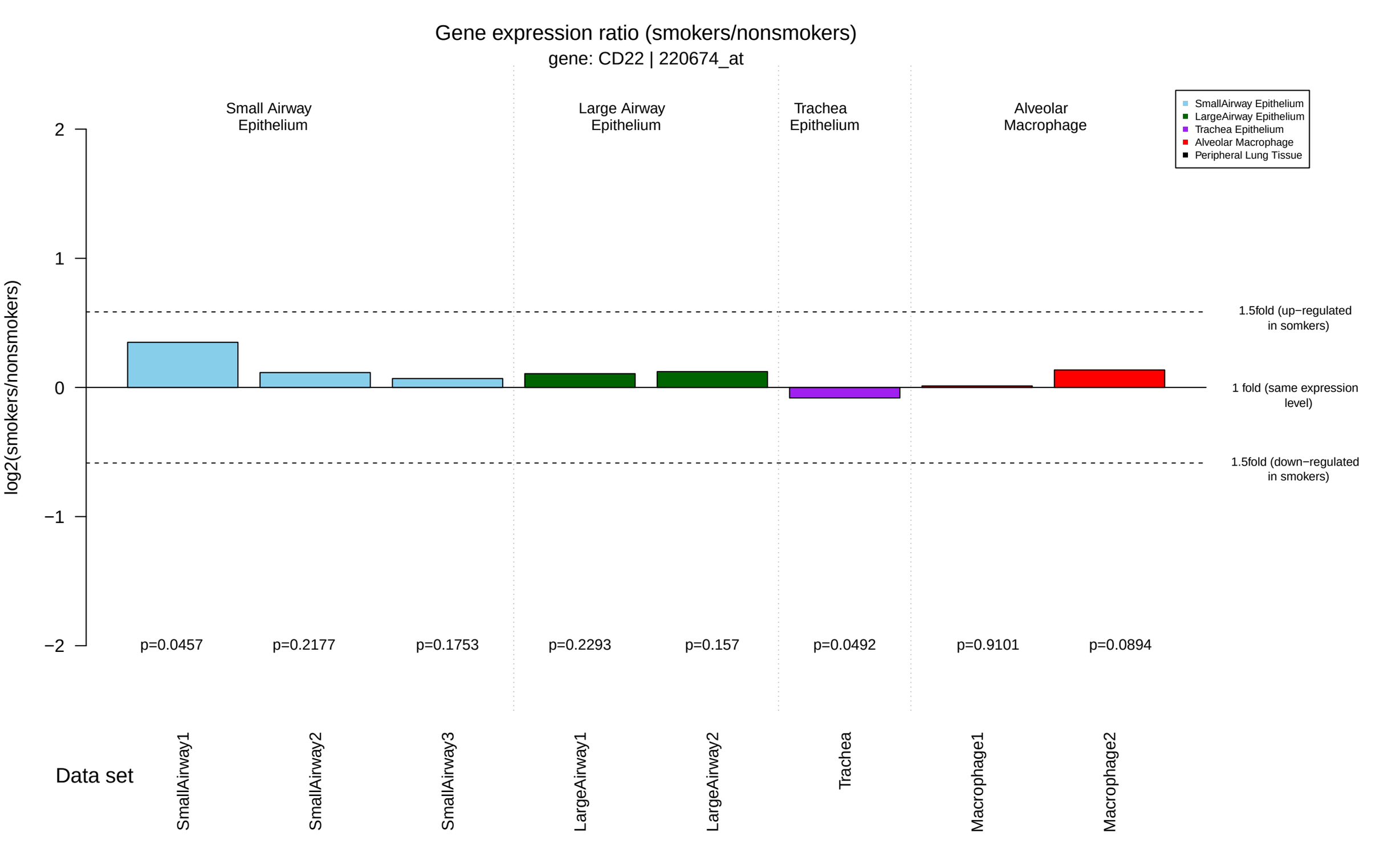
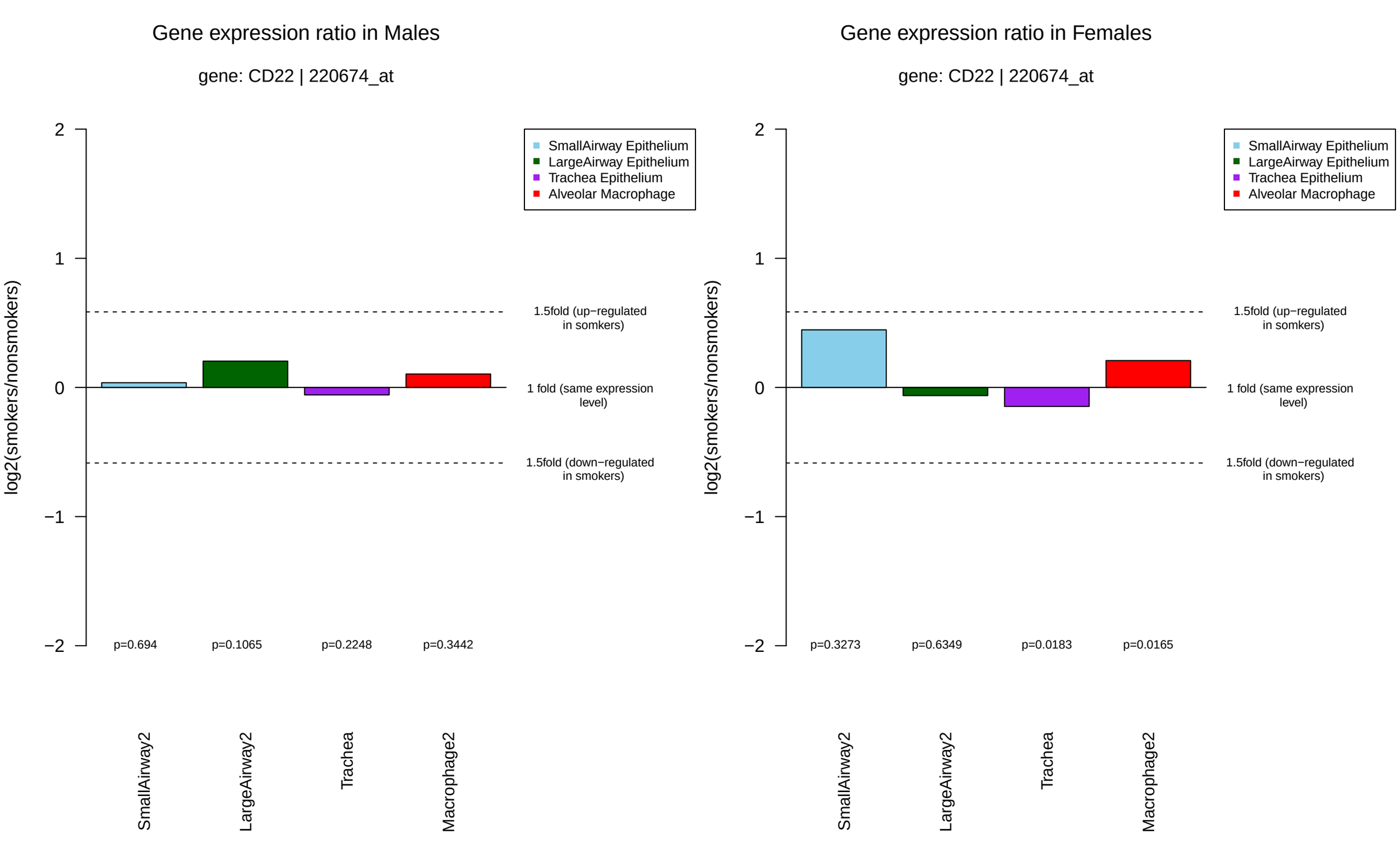
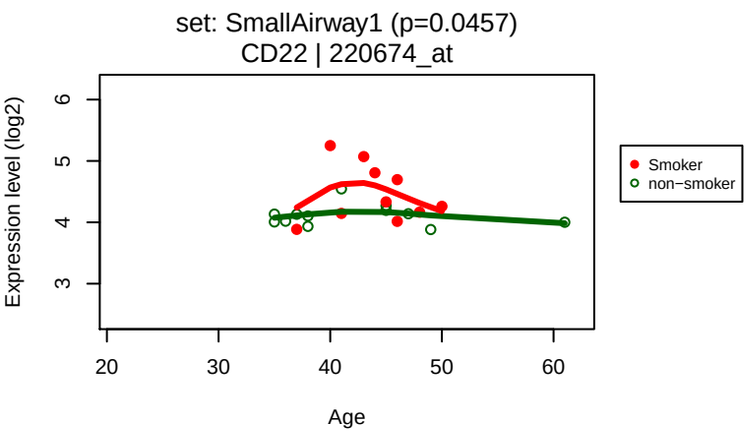
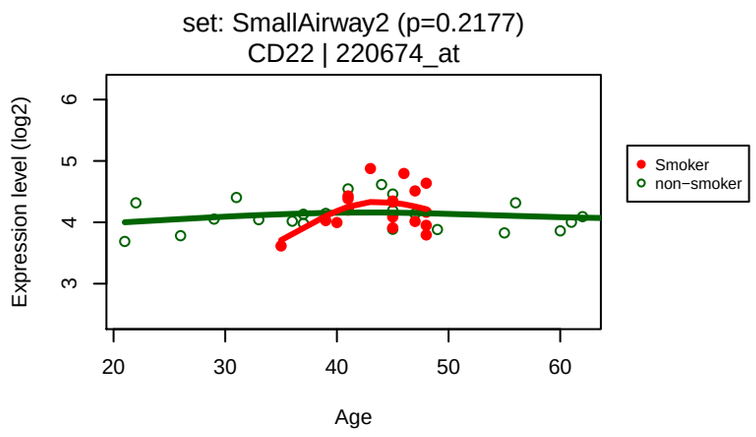
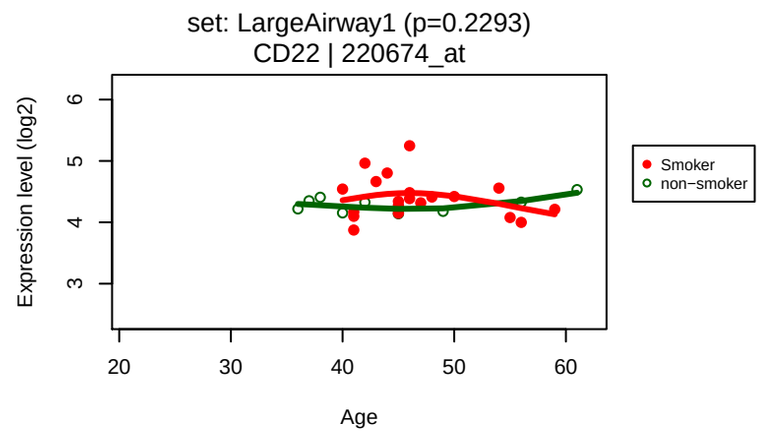

Gene expression of smokers and nonsmokers
gene: CD22 | 220674_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0457
p=0.2177
p=0.1753
p=0.2293
p=0.157
p=0.0492
p=0.9101
p=0.0894
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

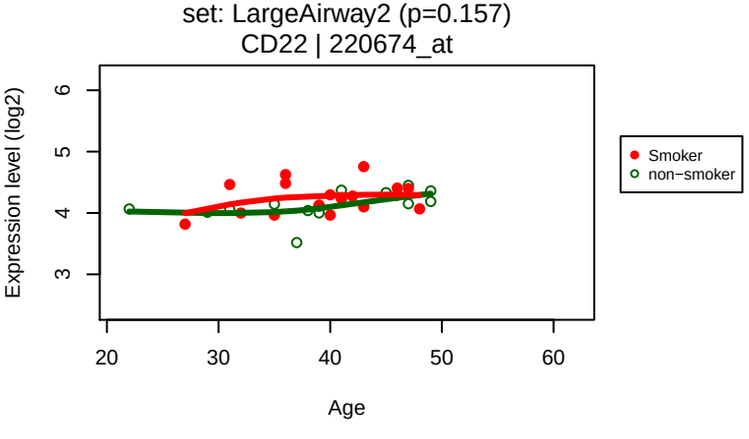

Gene expression ratio (smokers/nonsmokers)
gene: CD22 | 220674_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0457
p=0.2177
p=0.1753
p=0.2293
p=0.157
p=0.0492
p=0.9101
p=0.0894
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

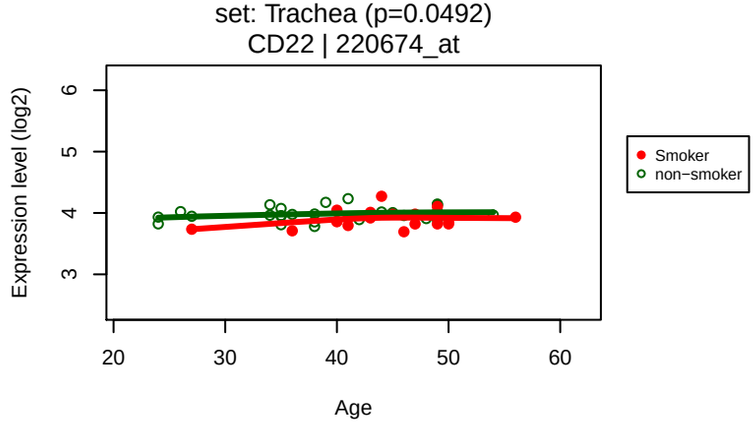

Gene expression ratio in Males
gene: CD22 | 220674_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.694
p=0.1065
p=0.2248
p=0.3442
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CD22 | 220674_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3273
p=0.6349
p=0.0183
p=0.0165
SmallAirway2
LargeAirway2
Trachea
Macrophage2

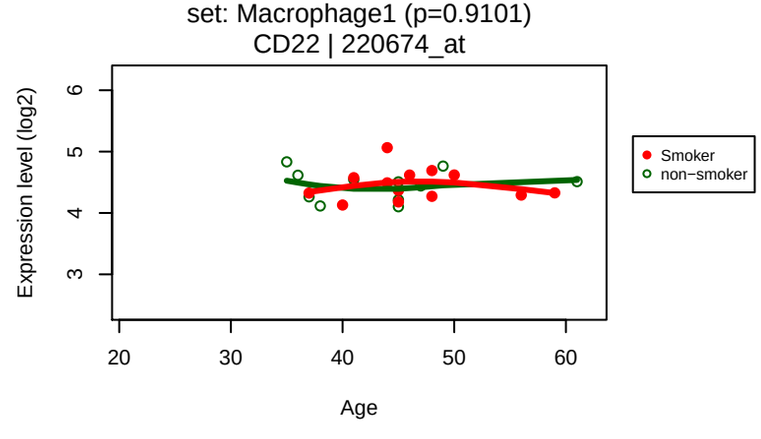

set: SmallAirway1 (p=0.0457)
CD22 | 220674_at
Expression level (log2)
Age
Smoker
non−smoker

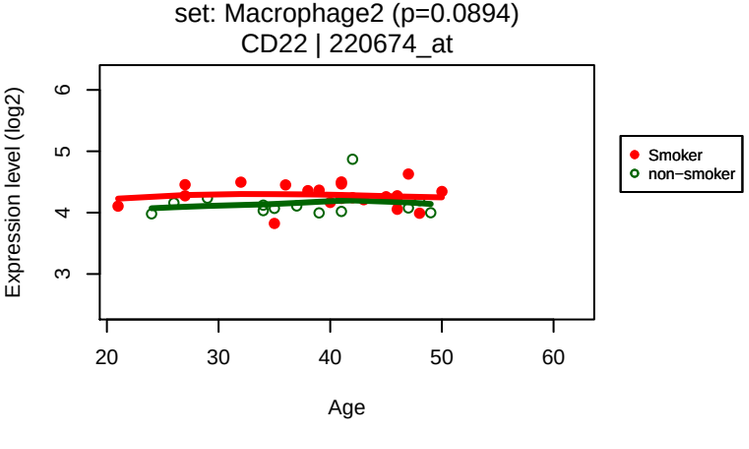

set: SmallAirway2 (p=0.2177)
CD22 | 220674_at
Expression level (log2)
Age
Smoker
non−smoker

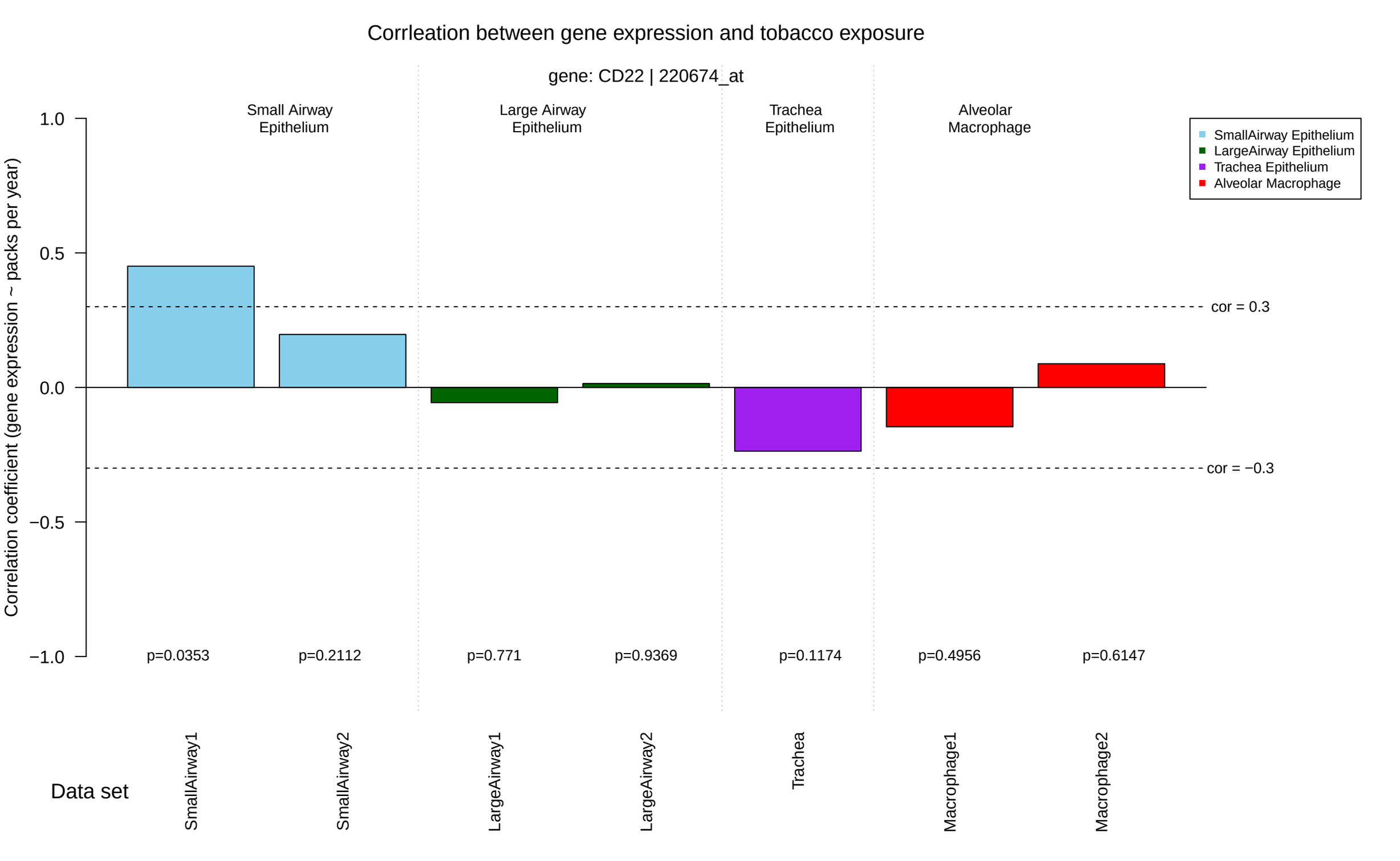

set: LargeAirway1 (p=0.2293)
CD22 | 220674_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.157)
CD22 | 220674_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0492)
CD22 | 220674_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9101)
CD22 | 220674_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0894)
CD22 | 220674_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CD22 | 220674_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0353
p=0.2112
p=0.771
p=0.9369
p=0.1174
p=0.4956
p=0.6147
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2